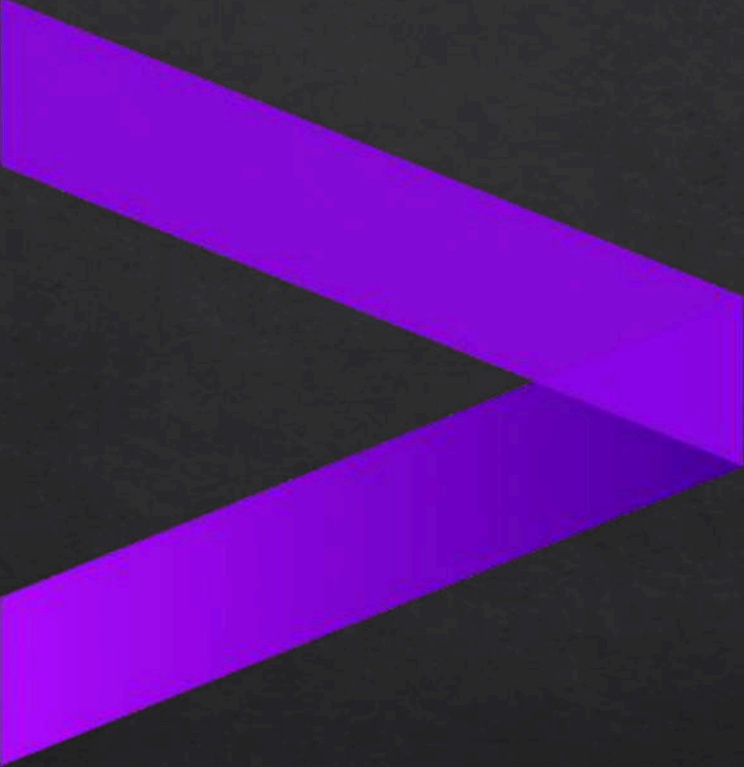


Nano-Enigma:

Uncovering the Secrets Within eFuse Memories

Finding and Reading Secrets from eFuse-Based Memories



Introduction

Embedded Systems Security Testing Team

Prague – Czech Republic

Martin Petran

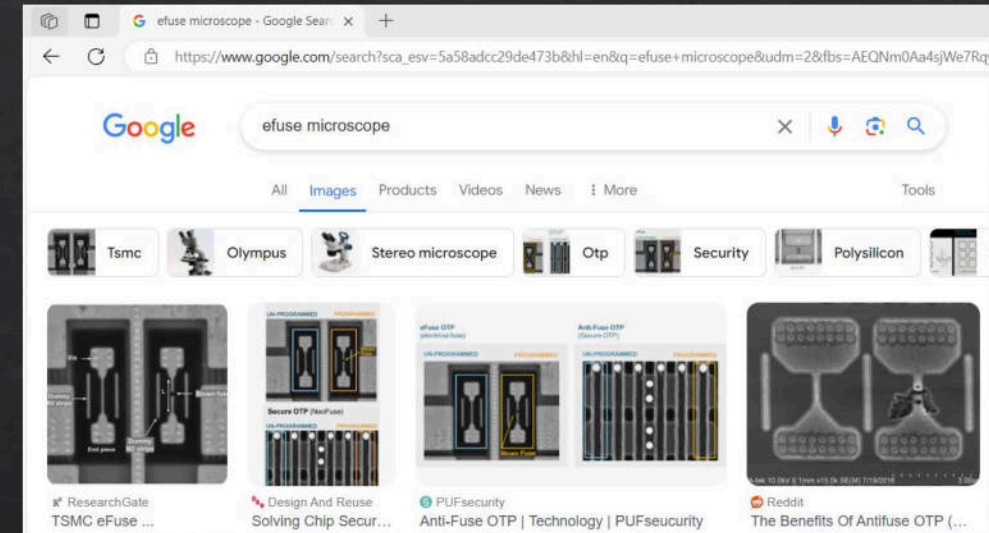
Michal Grygarek

Hayyan Ali

Overview

Answering basic questions:

- What is an eFuse and what are the benefits of using it for security?
- What exactly are these used for in security context?
- How are these protected from reading?

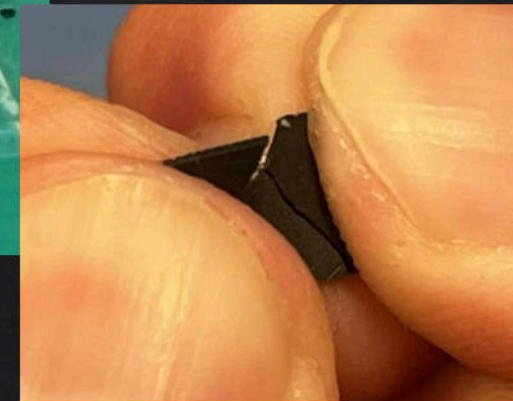
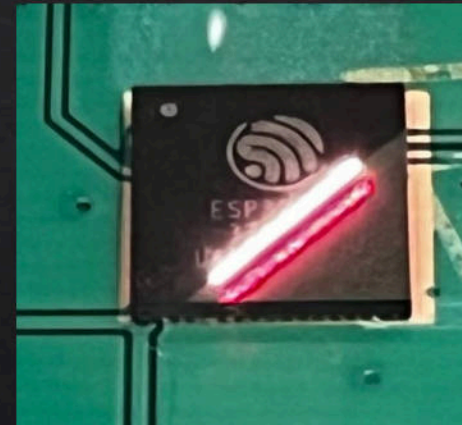
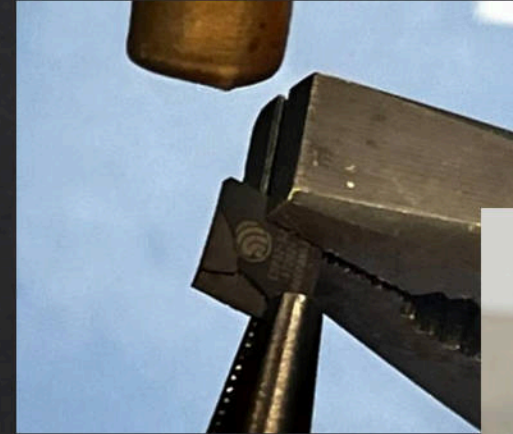


Execution of the attack



Sample preparation - Decapsulation

- Thermal – (100 USD)
 - Brute force, High risk of damage
- Chemical decapsulation - (~100 – 10.000 USD)
 - Gentle technique, Good results, Adequate equipment is necessary
- Laser etching - (~1.000 – 10.000 USD)
 - High risk of burning DIE, Good combination with the chemical method
- Plasma / ION etching - (~100.000 - 1.000.000 USD)
 - Gentle and precise technique

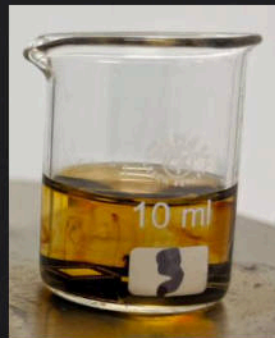


Chemical Decapsulation

- We chose Chemical Decapsulation due to its availability
 - In combination with laser etching, it delivers excellent results.
 - It completely removes all housing.
 - The quality of the result is repeatable.



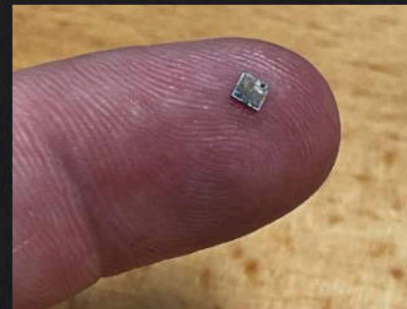
H_2SO_4 - 96%



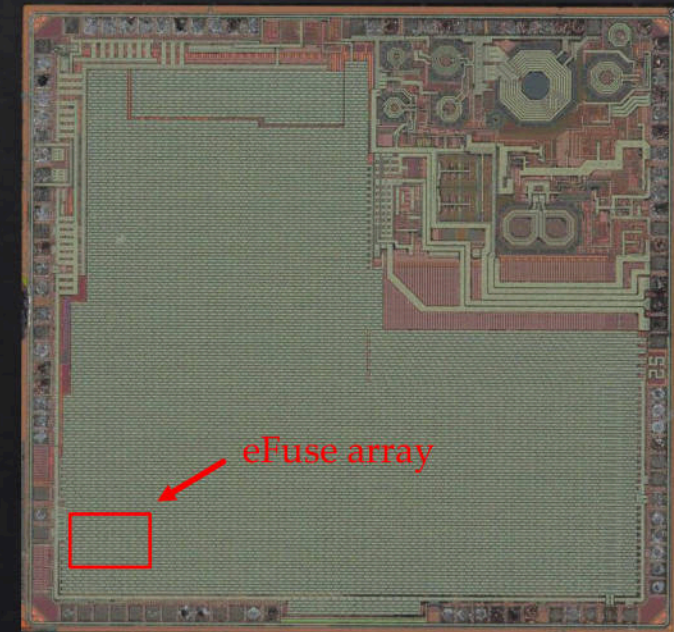
Evolving gas
 SO_x

Heat up to
 250°F

Rinsing residues



Protective equipment
is necessary

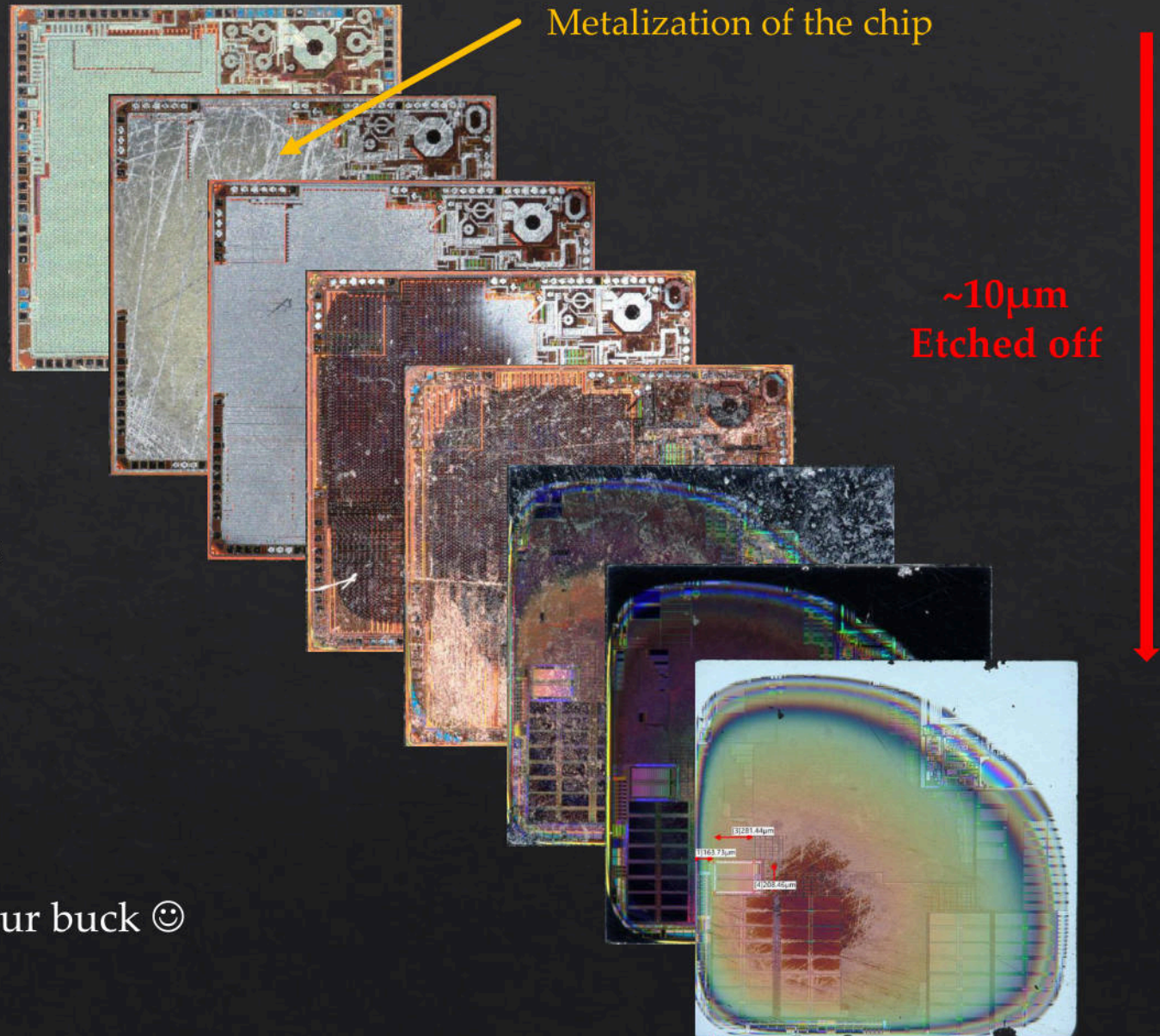


Perfectly clean DIE

Delayering

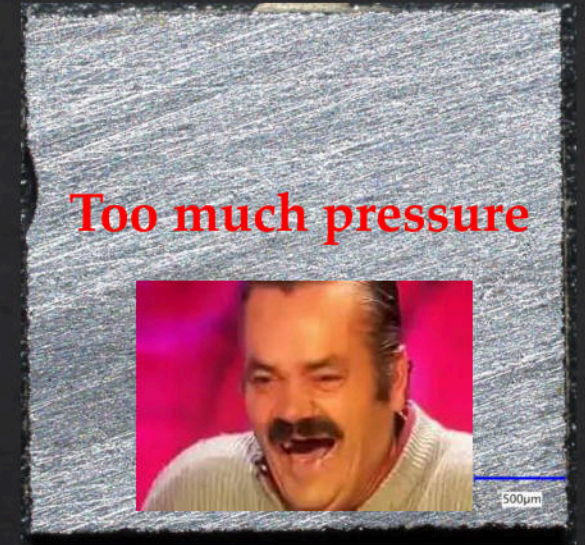
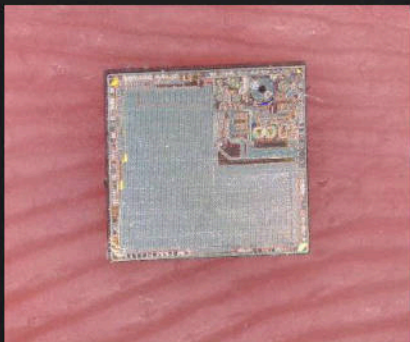
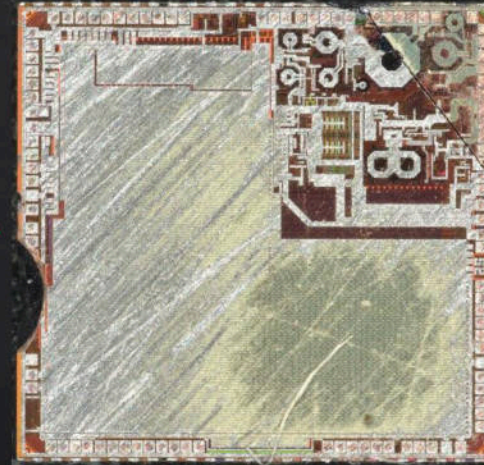
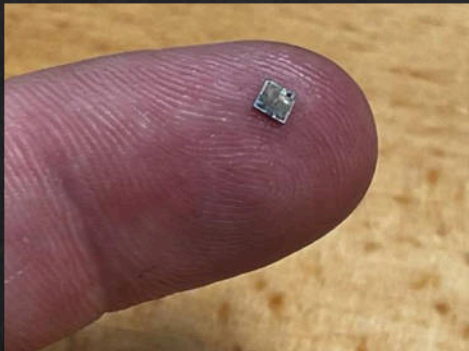
- Mechanical (100 USD)
- Chemical (250 USD)
- Laser ablation (~50.000 USD)
- FIB / Plasma (~2.000.000 USD)
- We use mechanical delayering because
 - Easy accessibility
 - Cheapest
 - Minimum equipment required

Mechanical way - a lot of bang for your buck ☺



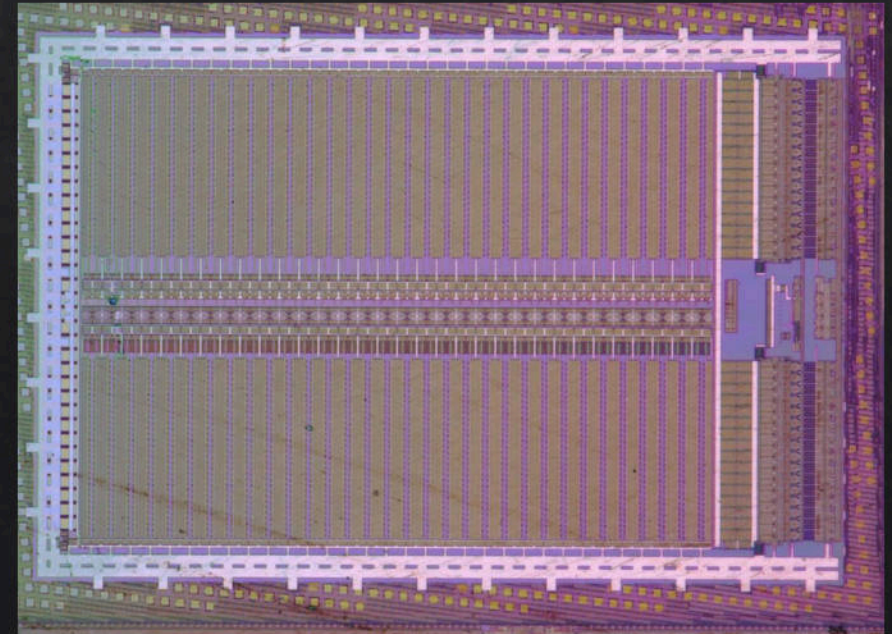
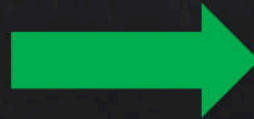
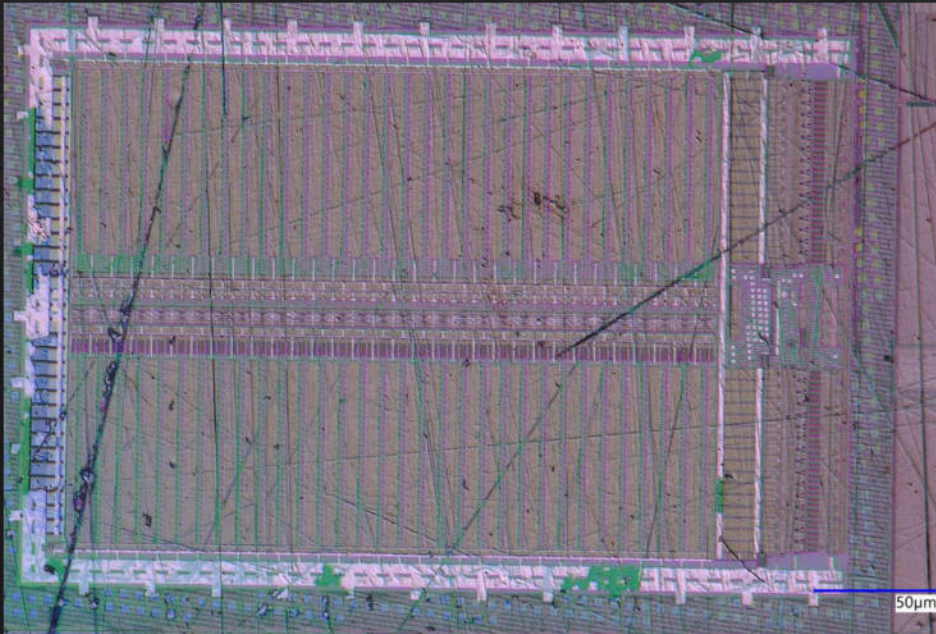
Delayering – wetstone

- It is necessary to have an abrasive surface with the highest possible grit size
- A capable worker prepares one sample within 60 min
- The skin on the fingers of one hand is enough for delayering 3 ESP32 chips.



Delayering – polishing

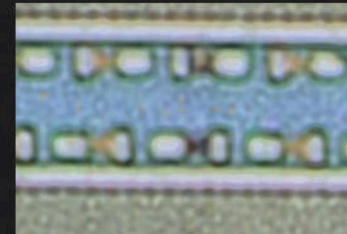
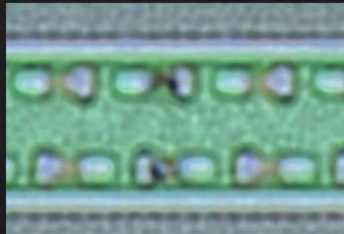
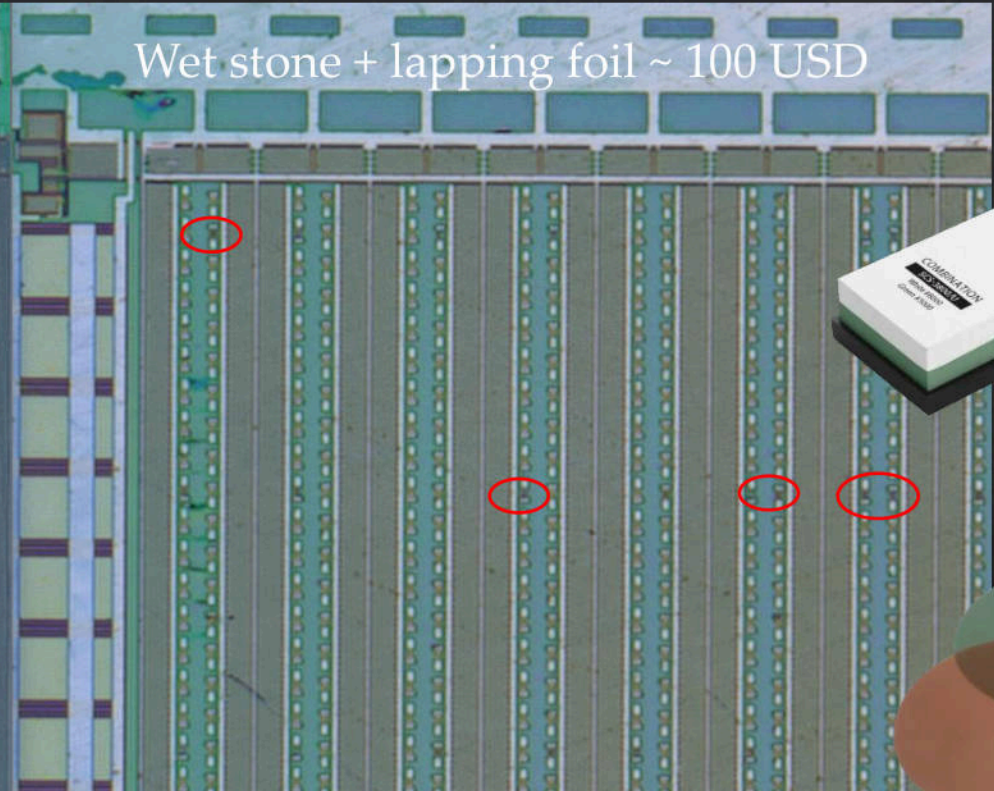
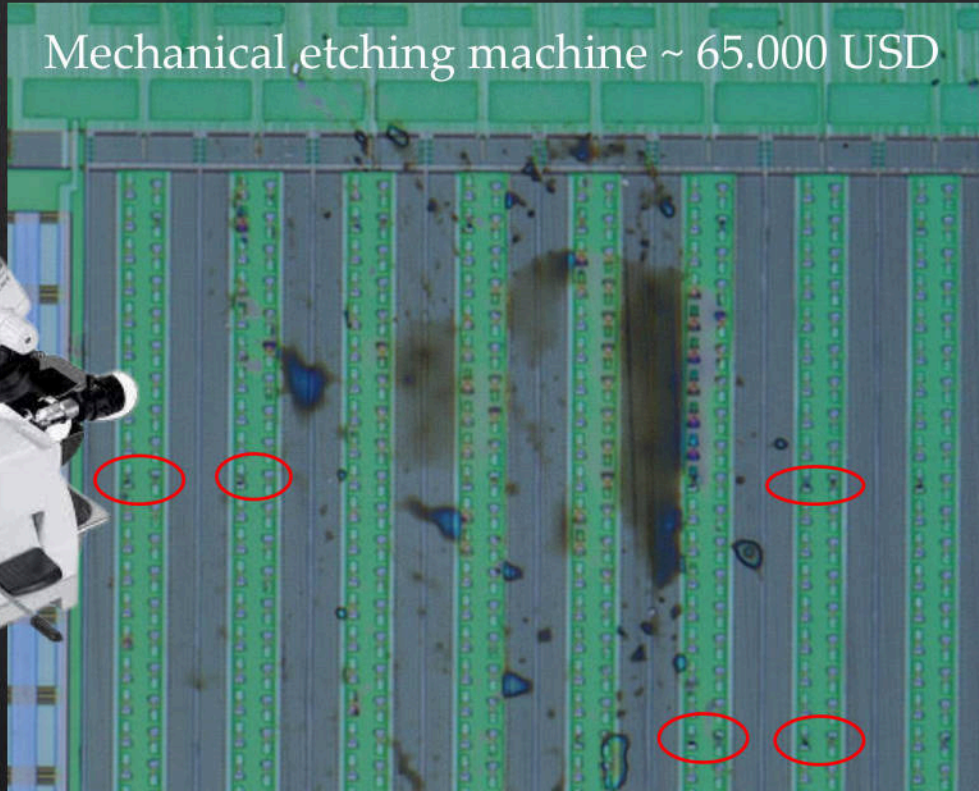
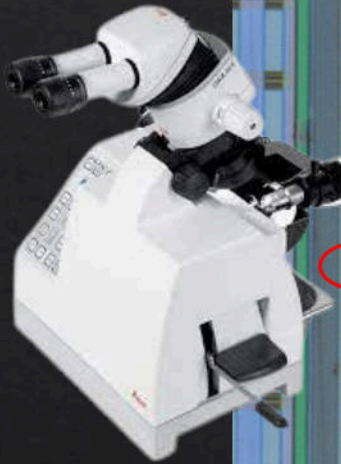
- Initial etching on a wet stone with a grit of 8000 JIS ($2\mu\text{m}$)
- The last phase needs to be finished on the diamond lapping foil ($<0,5\mu\text{m}$)



Delayering – methode comparing

Mechanical etching machine ~ 65.000 USD

Wet stone + lapping foil ~ 100 USD



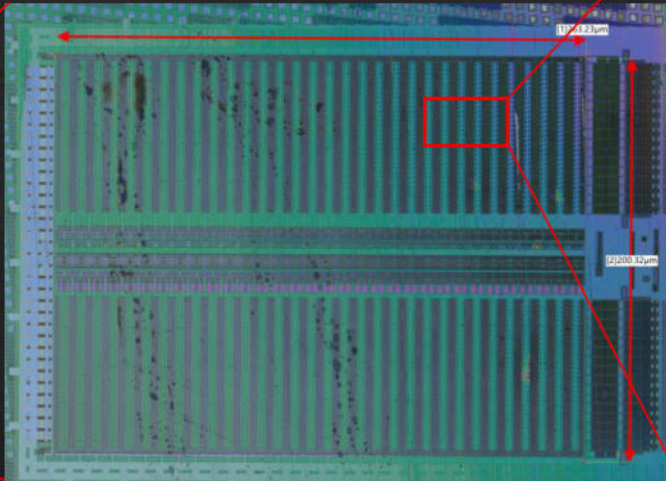
Localization of the On-chip eFuse field

- We evaluated the eFuse bank identity with the ESP32 S2 / S3 / C6 model.
- eFuse is essentially a fuse

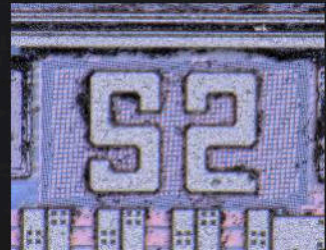
Magnification 80x



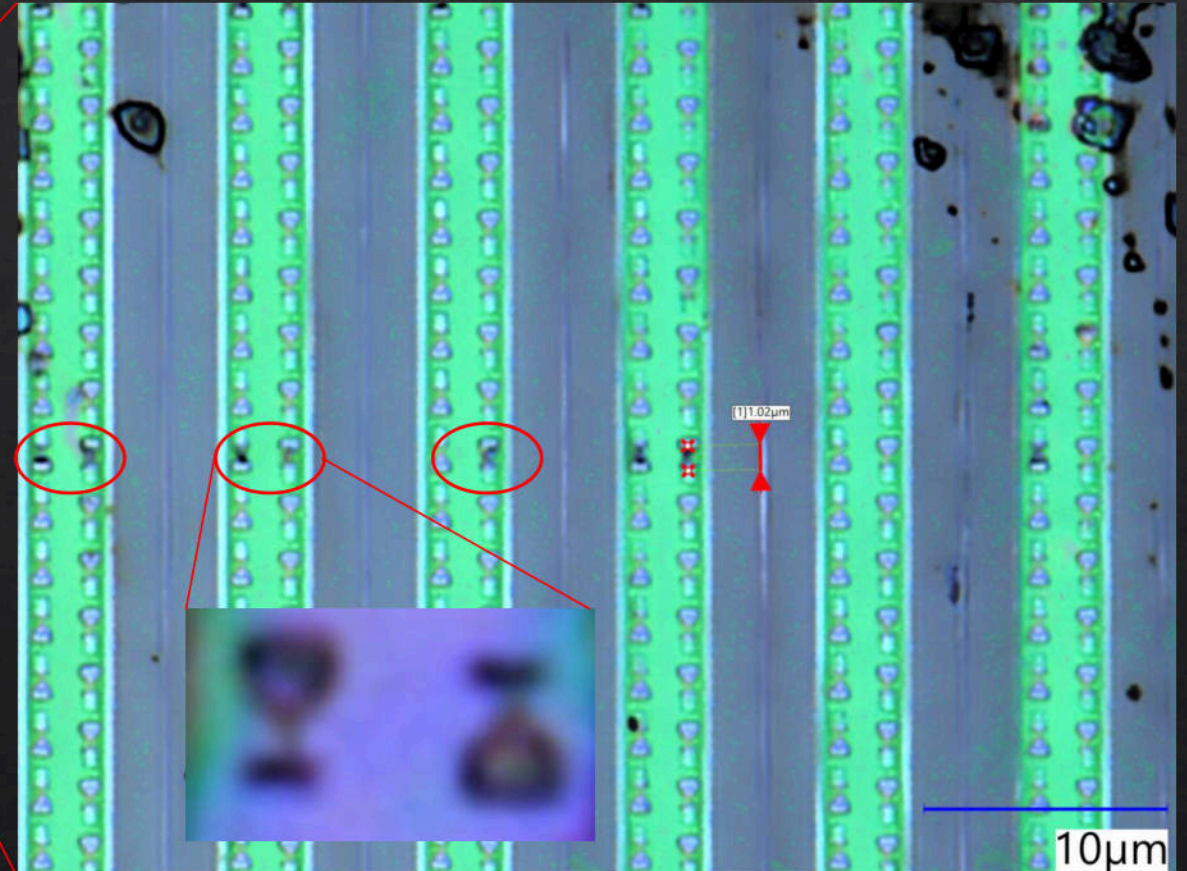
Magnification 700x



View of the complete 4096-bit eFuses bank
PROM bits



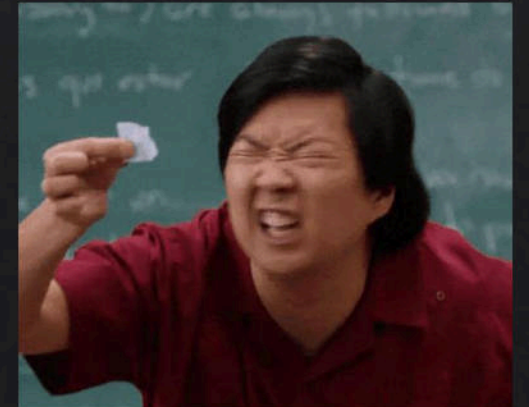
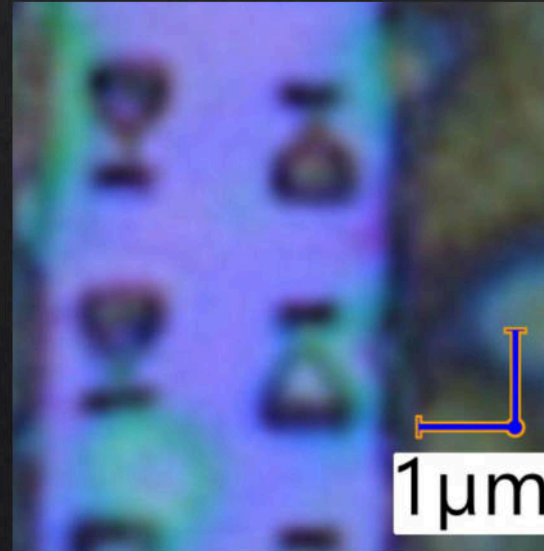
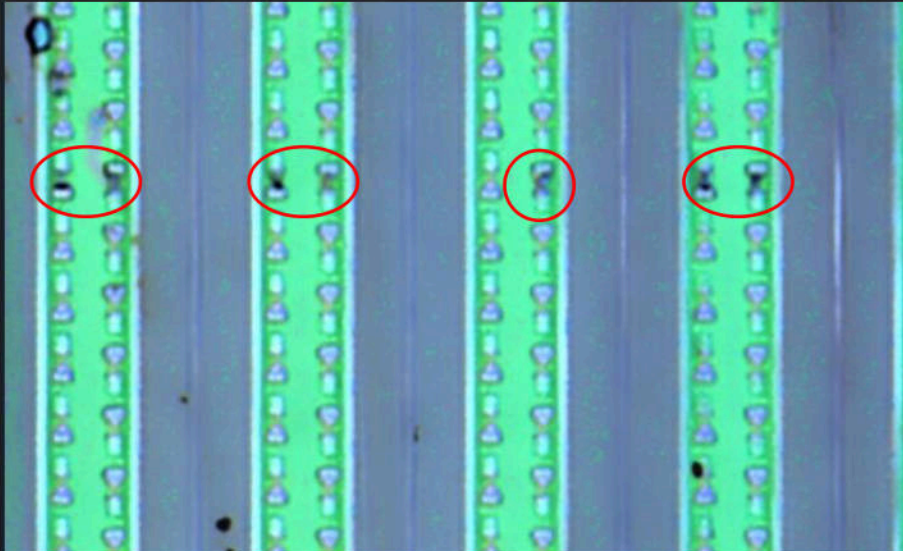
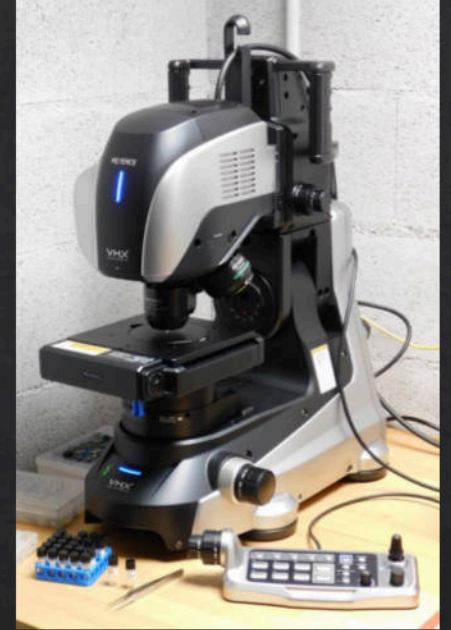
Magnification 2000x



Detail of individual eFuses, here we can see that optical microscopy no longer provides the required level of detail

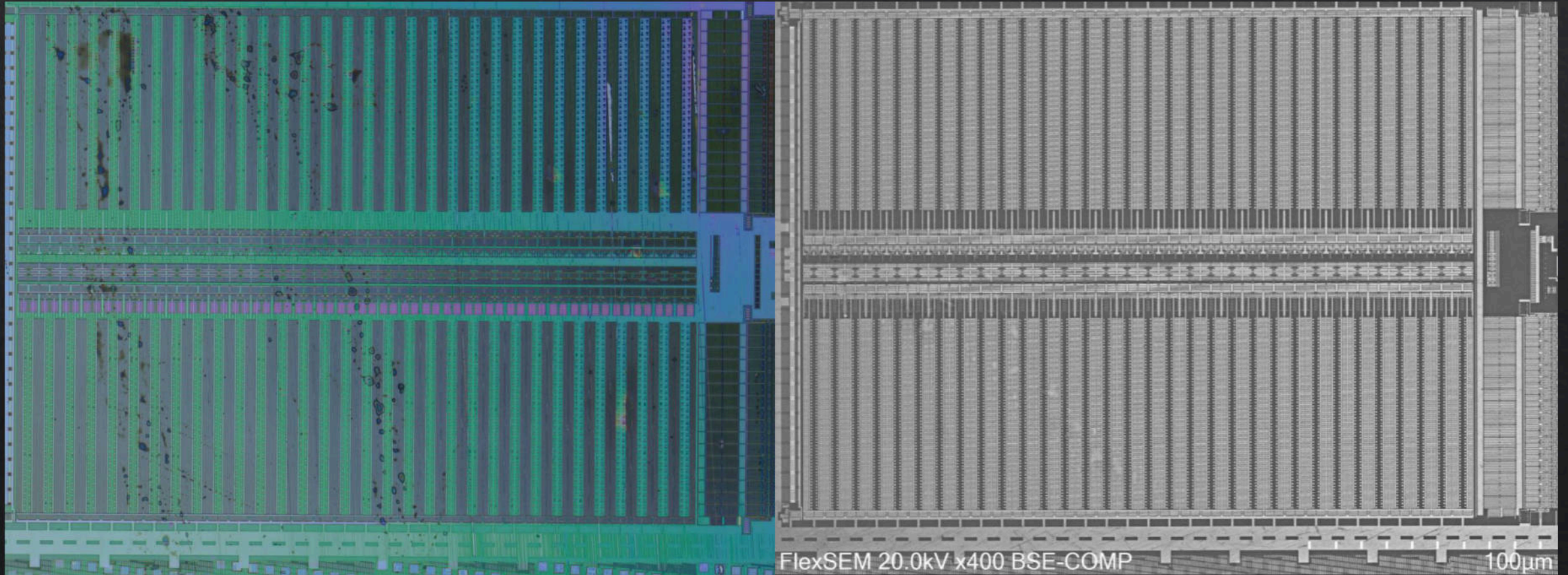
Optical microscopy observing - limits

- Used equipment – Keyence VHX-7000
 - Maximum resolution ~ 300nm (~2000x magnification) – Physical limits
- Sufficient for locating the eFuse bank
- On a well-prepared sample, even burned eFuses can be seen
 - The resolution is insufficient for reliable state reading of eFuses
 - A human hair is 100 times bigger than an eFuse



We need Scanning Electron Microscopy

- We need more advanced imaging technique
- SEM provides higher resolution

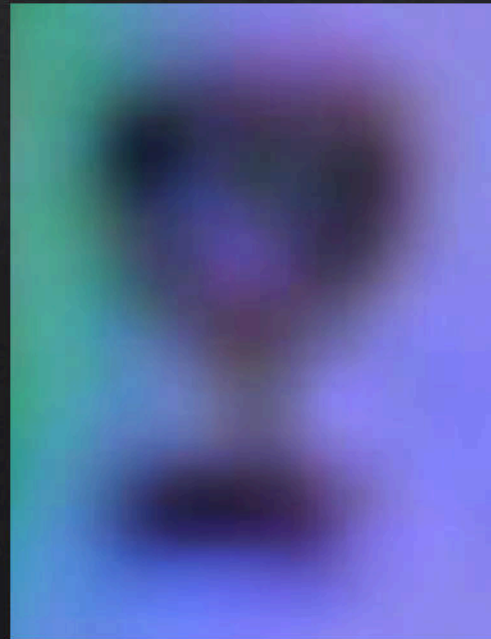


Optical vs. Scanning Electron Microscopy

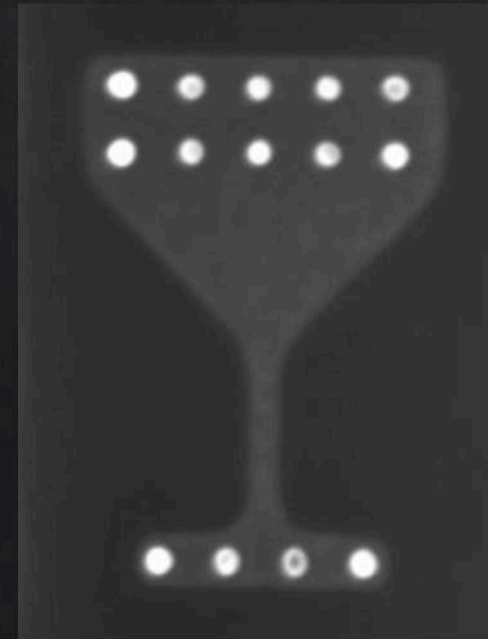


Optical
Resolution ~300nm
Magnification 2000x

- SEM provides ~100 times higher resolution
- Structural changes are more observable
- eFuse is 100 x smaller than a human hair
- The level of detail is significant

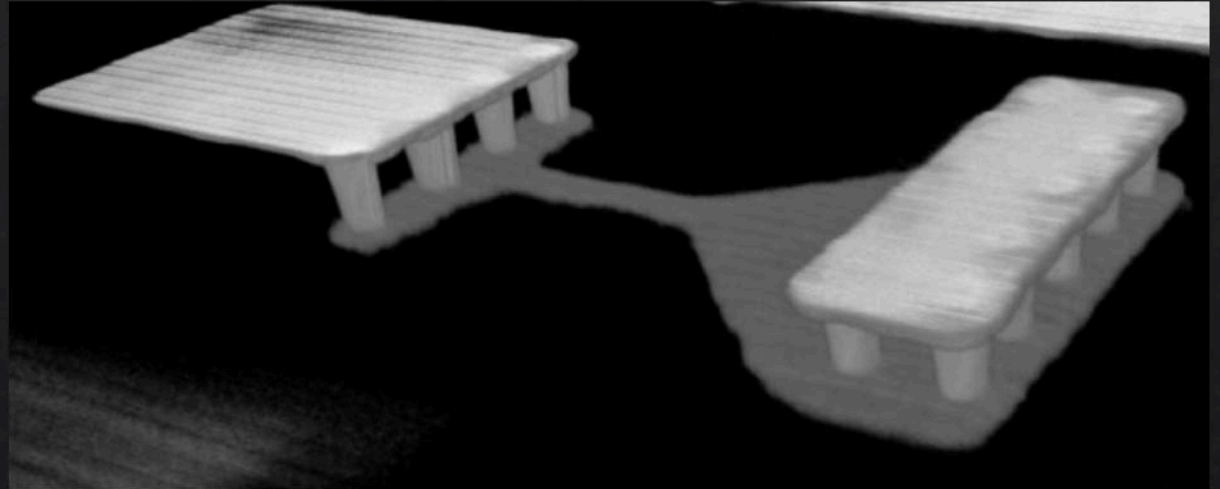
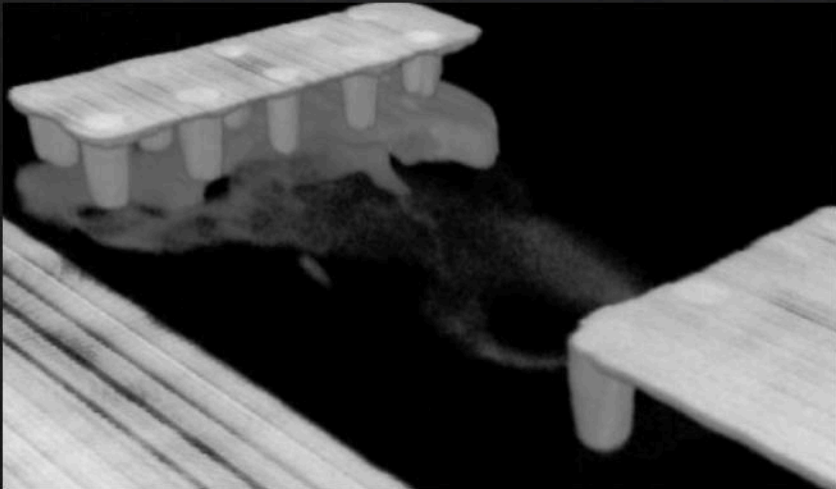
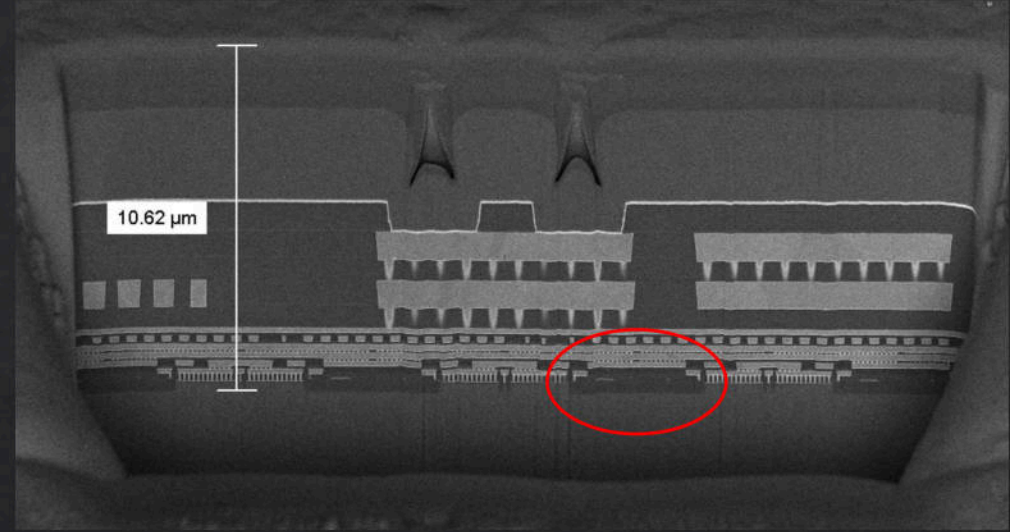


1 μ m



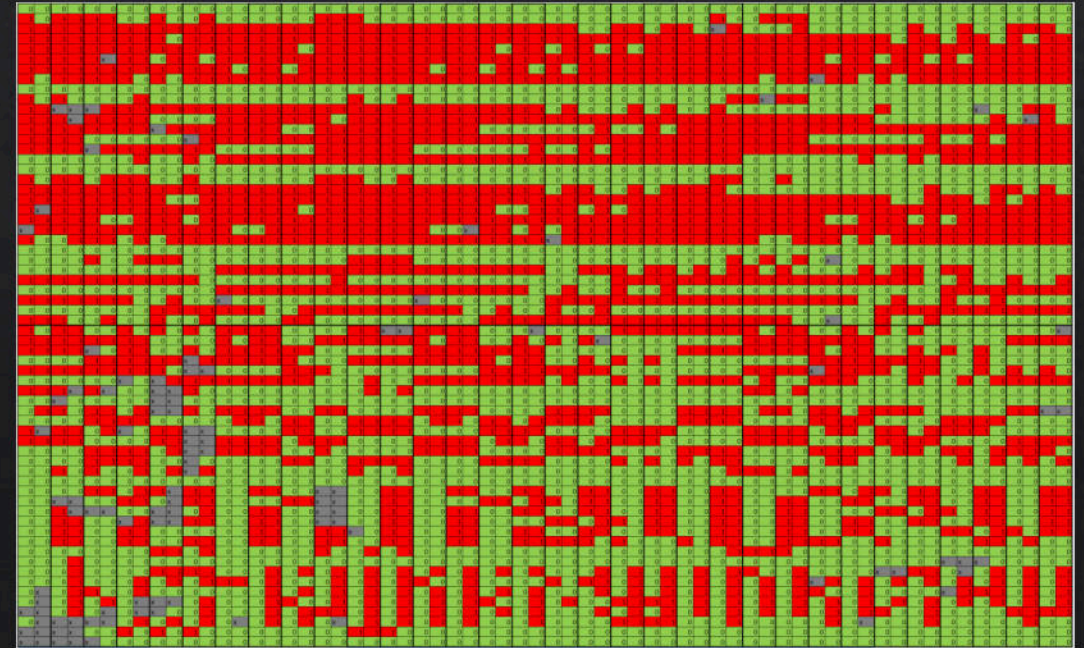
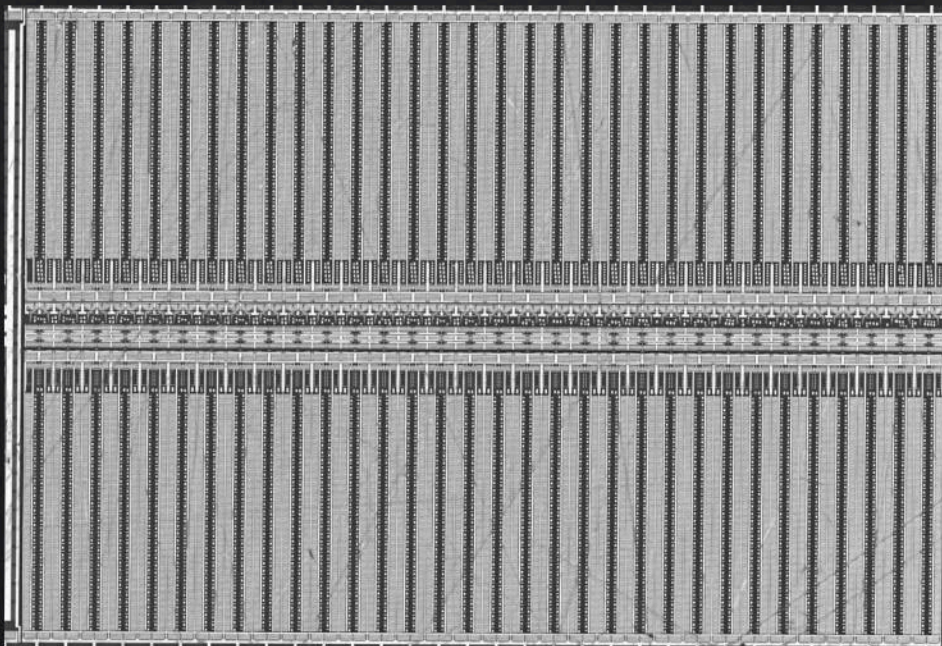
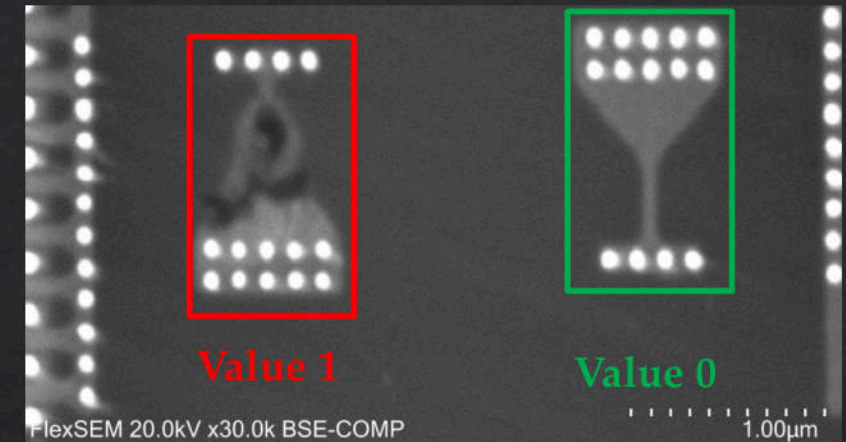
SEM
Resolution ~5nm
Magnification 100.000x

Electron microscopy observe details



eFuses data extraction – „visually reading“

- Extremely time consuming, intense focus
 - Reading a 4096-bit bank takes ~10 hours of focused work
- In the initial stages, about ~20kb of eFuses were read visually ☺



Manually transcribed into Excel

Memory layout evaluation

- **What we know?**
- The documentation mentions:
 - Memory is organized in blocks.
 - Total of 11 eFuse blocks
 - Blocks 0-2 are used for system purposes
 - The rest of the blocks are user programmable
 - There are 4 backups of BLOCK0 and BLOCK1-10 are using RS coding scheme for automatic error correction (adds 12 bytes to each block).
- **What we want to know?**
 - How is the eFuse array physically mapped to logical blocks?

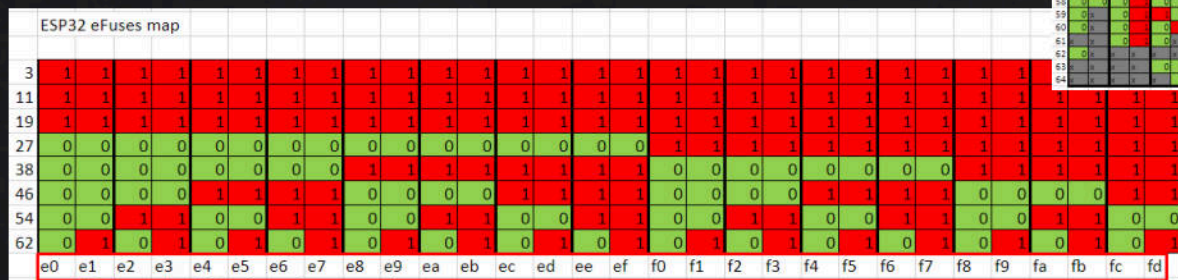
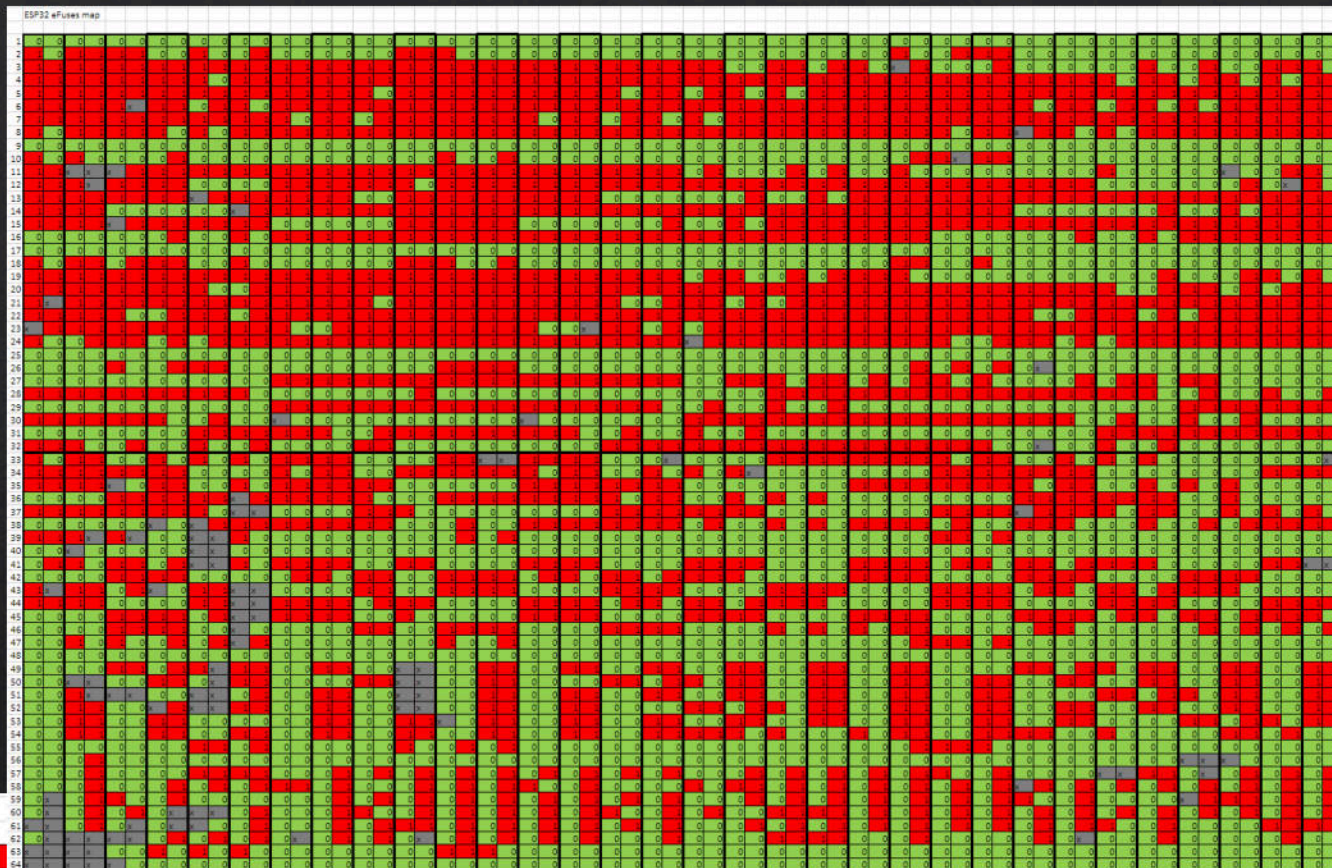
Memory layout evaluation

Manual Experiment #2:

Burn all user writeable eFuse bytes with identical pattern.

Bingo ...

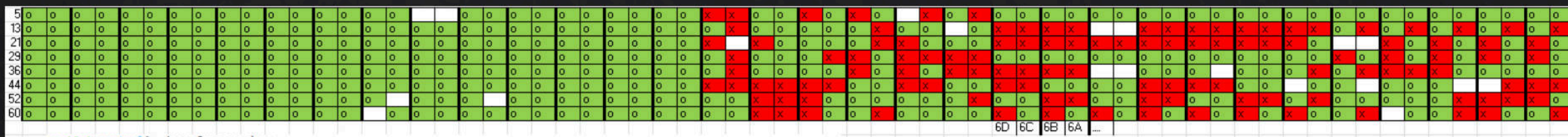
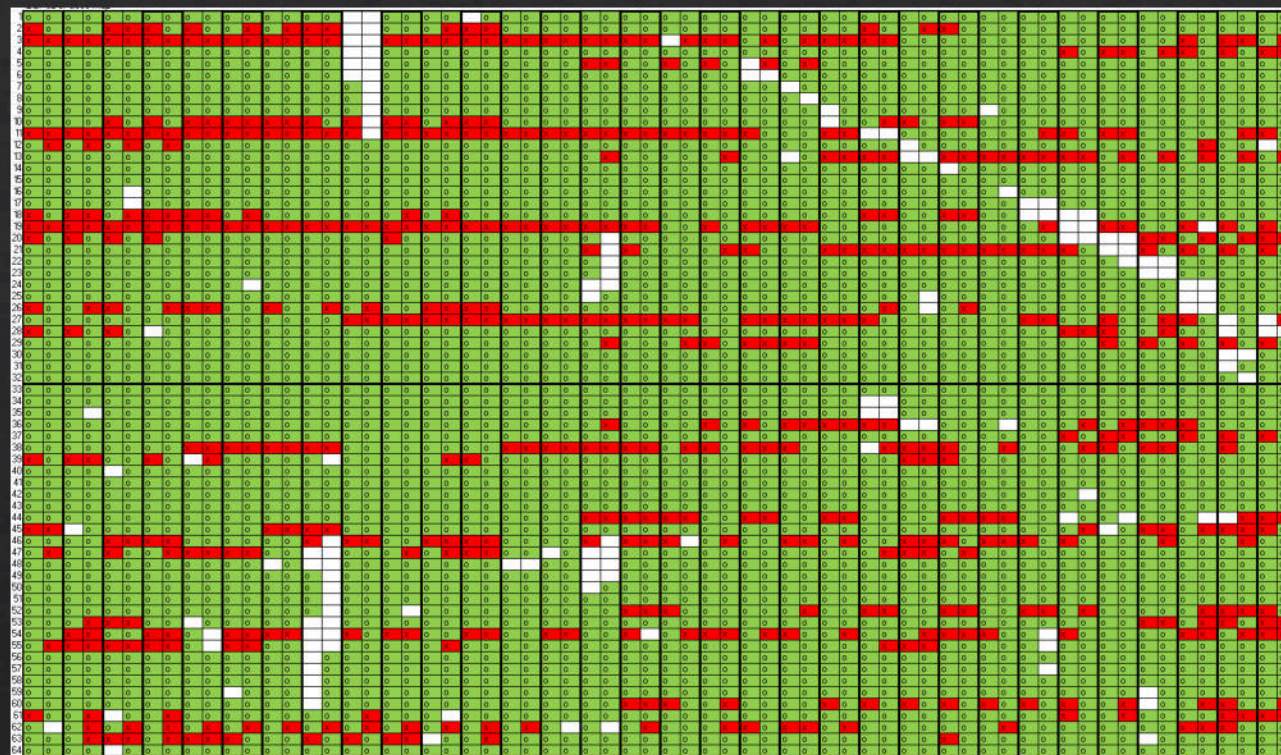
```
espefuse.py v4.7-dev
Connecting.....
Detecting chip type... Unsupported detection protocol, switching and trying again...
Detecting chip type... ESP32-S2
BLOCK0 ( [0 ] read_reg: 00000000 00000000 00000000 00000000 00000000 00000000
MAC_SPI_8M_0 (BLOCK1 ) [1 ] read_reg: 4e9edff2 0000a076 00000000 00000000 00000000 00000000
BLOCK_SYS_DATA (BLOCK2 ) [2 ] read_reg: a5cd9406 cb53e8ac 0f379edc 2c4970a6 145c882d 401028a2 0e232970 0604491d
BLOCK_USR_DATA (BLOCK3 ) [3 ] read_reg: fcfdfef7 f8f9fafb f4f5f6f7 f0f1f2f3 ecdeeeef e8e9eaeb e4e5e6e7 e0e1e2e3
BLOCK_KEY0 (BLOCK4 ) [4 ] read_reg: fcfdfef7 f8f9fafb f4f5f6f7 f0f1f2f3 ecdeeeef e8e9eaeb e4e5e6e7 e0e1e2e3
```



Memory layout evaluation

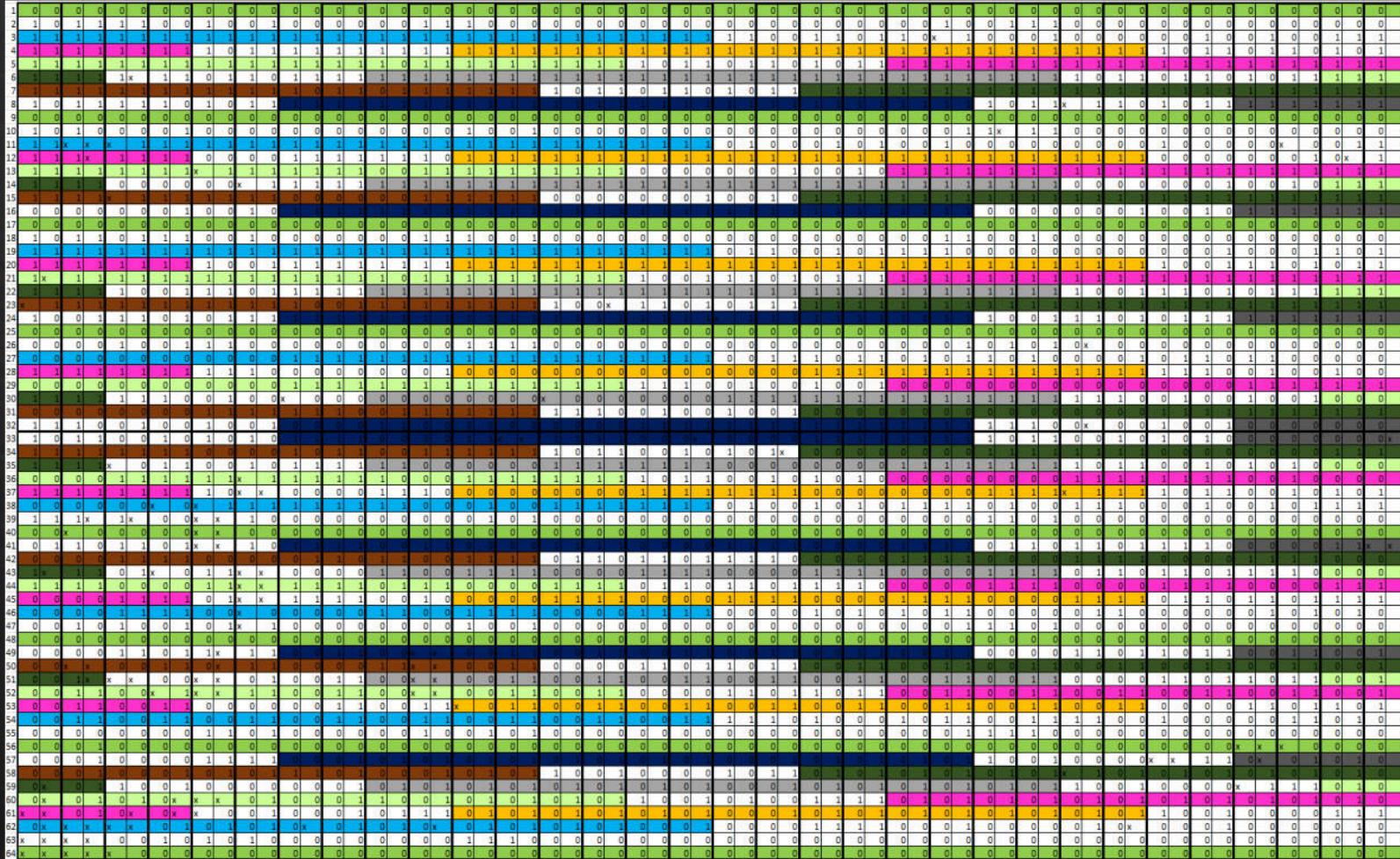
Manual Experiment #3:

Burn unique values to BLOCK3 and BLOCK5 (0x6a 0x6b 0x6c 0x6d ...).

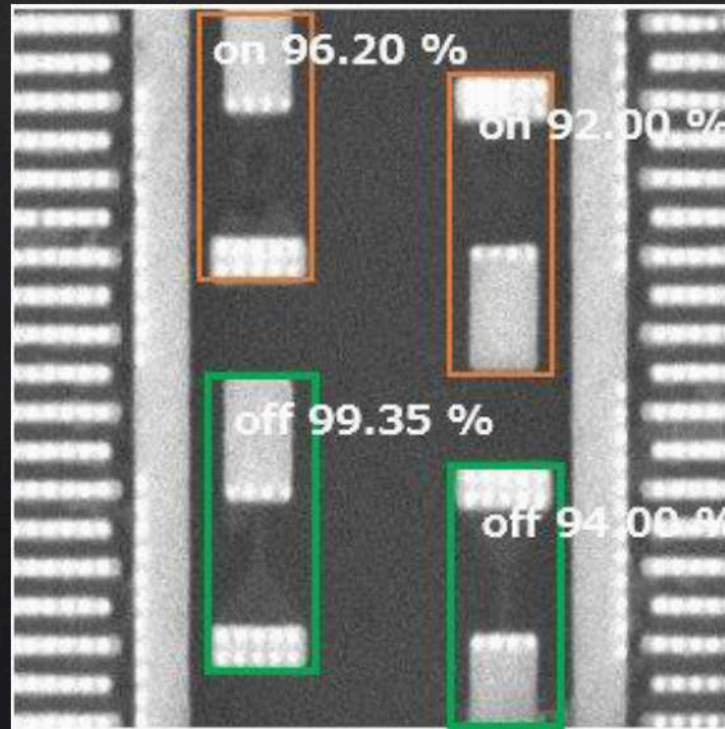


Memory layout evaluation

At this point we know what is what ...



Automating the Process



Conclusion

- eFuse-based memories do not provide any confidentiality assurance
- Readability protection methods such as layering increase effort but do not prevent reading
- If you can rent a microscope, it is not that expensive to perform such attack
- Espressif has great team handling submissions for potential security issues
- Integrity assurance of the eFuse-based memories ... to be continued 😊

Q & A